

# Integrating Open OnDemand & Open XDMoD

Tutorial presented at  
Global OnDemand Conference (GOOD) 2025

Dori Sajdak, Senior Systems Administrator  
University at Buffalo Center for Computational Research



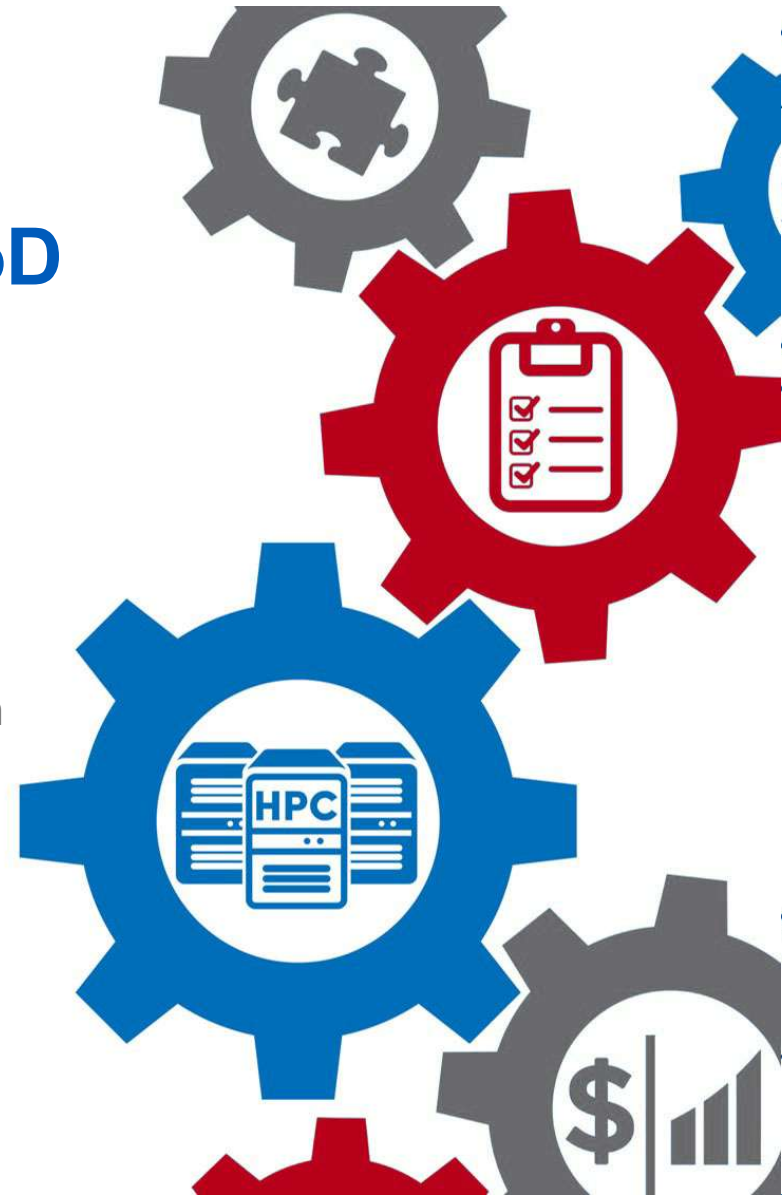
**Ohio Supercomputer Center**

An **OH·TECH** Consortium Member



University at Buffalo

Center for Computational Research





**Ohio Supercomputer Center**  
An **OH·TECH** Consortium Member

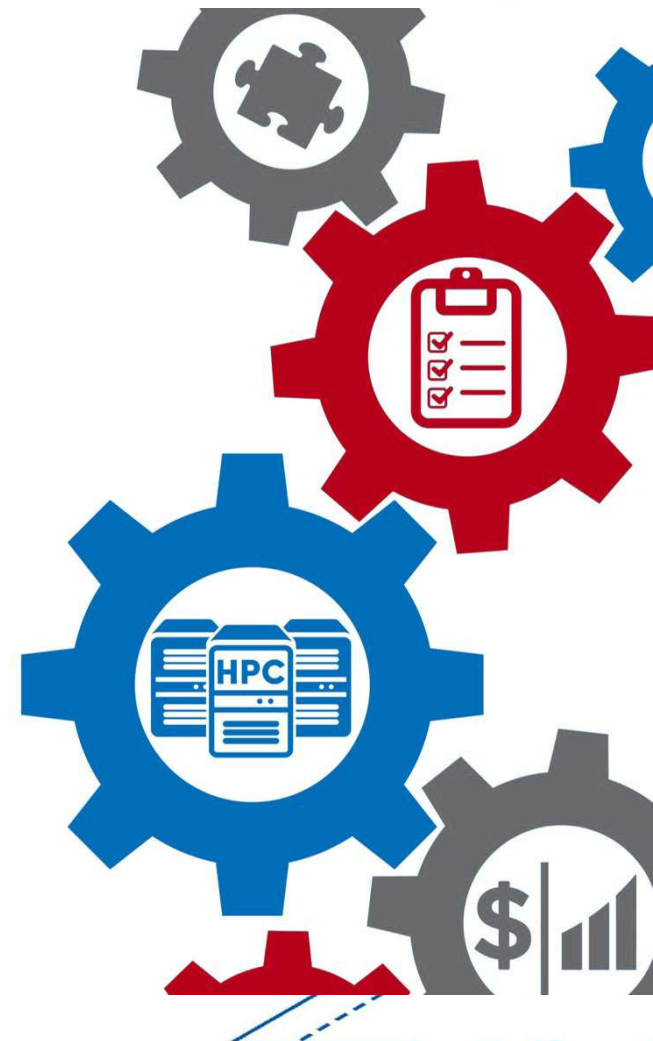


University at Buffalo

Center for Computational Research

## Agenda

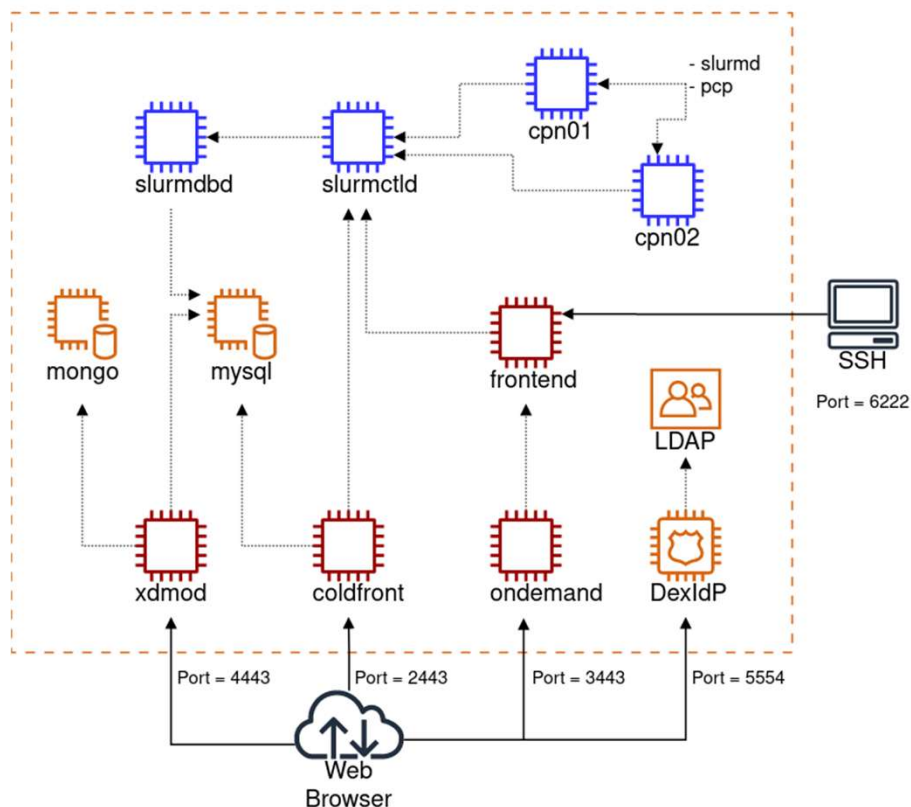
- Tutorial technology & walk through of the repo
- A bit about XDMoD
- XDMoD integration with OnDemand
- OnDemand integration with XDMoD
- Demo/tutorial
- Where to find us/get help/stay connected
- Q&A





# Tutorial Container Architecture

HPC Toolset Containers



**Requirements:** <https://github.com/ubccr/hpc-toolset-tutorial/edit/master/docs/requirements.md>

## Clone the Github Repo:

```
git clone https://github.com/ubccr/hpc-toolset-tutorial
cd hpc-toolset-tutorial
./hpcts start
```

\* The first time you do this, you'll be download ~25GB worth of containers from Docker Hub. This can take a long time depending on your network speeds. After downloaded, the containers are started, and services launched.

**WARNING!!! DO NOT run these containers on production systems. This project is for educational purposes only. The container images we publish for the tutorial are configured with hard coded, insecure passwords and should be run locally in development for testing and learning only.**





# HPC Toolset Tutorial Walk Through

<https://github.com/ubccr/hpc-toolset-tutorial>

Keep the applications page open for easy access to account  
usernames/passwords and portal URLs:

<https://github.com/ubccr/hpc-toolset-tutorial/blob/master/docs/applications.md>

We'll be following the XDMoD-OnDemand Integration Tutorial :

<https://github.com/ubccr/hpc-toolset-tutorial/blob/master/xdmod/README.md>

If you're following along, please do the first step of the tutorial now:

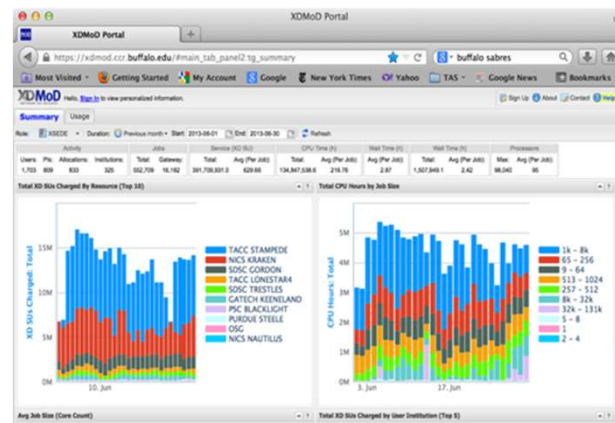
```
[hpcadmin@frontend ~]$ submit_jobs.sh
```



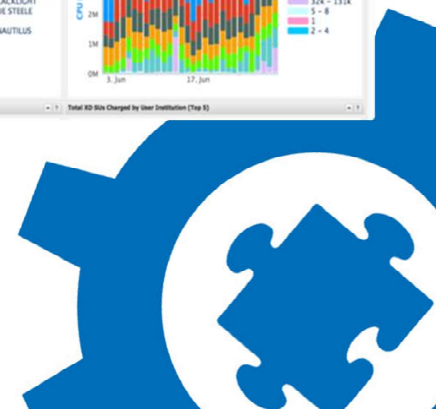


## XDMoD: Metrics on Demand

- **Comprehensive framework for CI system management**
  - HPC Jobs, Storage, Cloud, Networking, OnDemand, & other Gateway data
- **Understand and optimize resource utilization and performance**
  - Provide instantaneous and historical information on utilization
  - Measure Quality of Service of CI systems and applications
  - Measure and improve job and system level performance
  - Inform computing system upgrades and procurements
- **ACCESS XDMoD Tool**
  - Analytics Framework for ACCESS and NAIRR
- **Open XDMoD\*: Open Source version for CI centers**
  - Used to measure and optimize performance of CI centers
  - 500+ academic, governmental, & commercial installations worldwide



<https://open.xdmod.org/>





**Ohio Supercomputer Center**

An OH-TECH Consortium Member

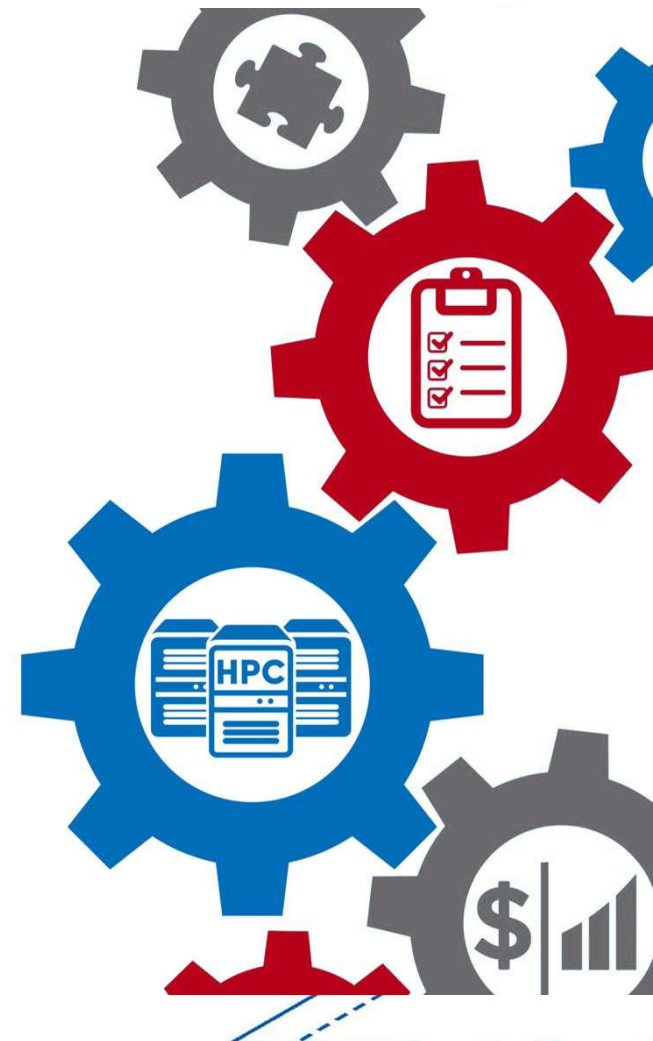


University at Buffalo

Center for Computational Research

## Open XDMoD – Usage & Performance Metrics

- Tool that aggregates job data & system performance metrics to inform system users, system staff & center decision makers
- Web portal providing job & system metrics, including utilization, quality of service metrics designed to proactively identify underperforming system hardware and software, and job level performance data for every job
- Role-based access to data with different levels of granularity, including job, user, or on a system-wide basis
- Ingest OnDemand logs into new OnDemand realm in XDMoD

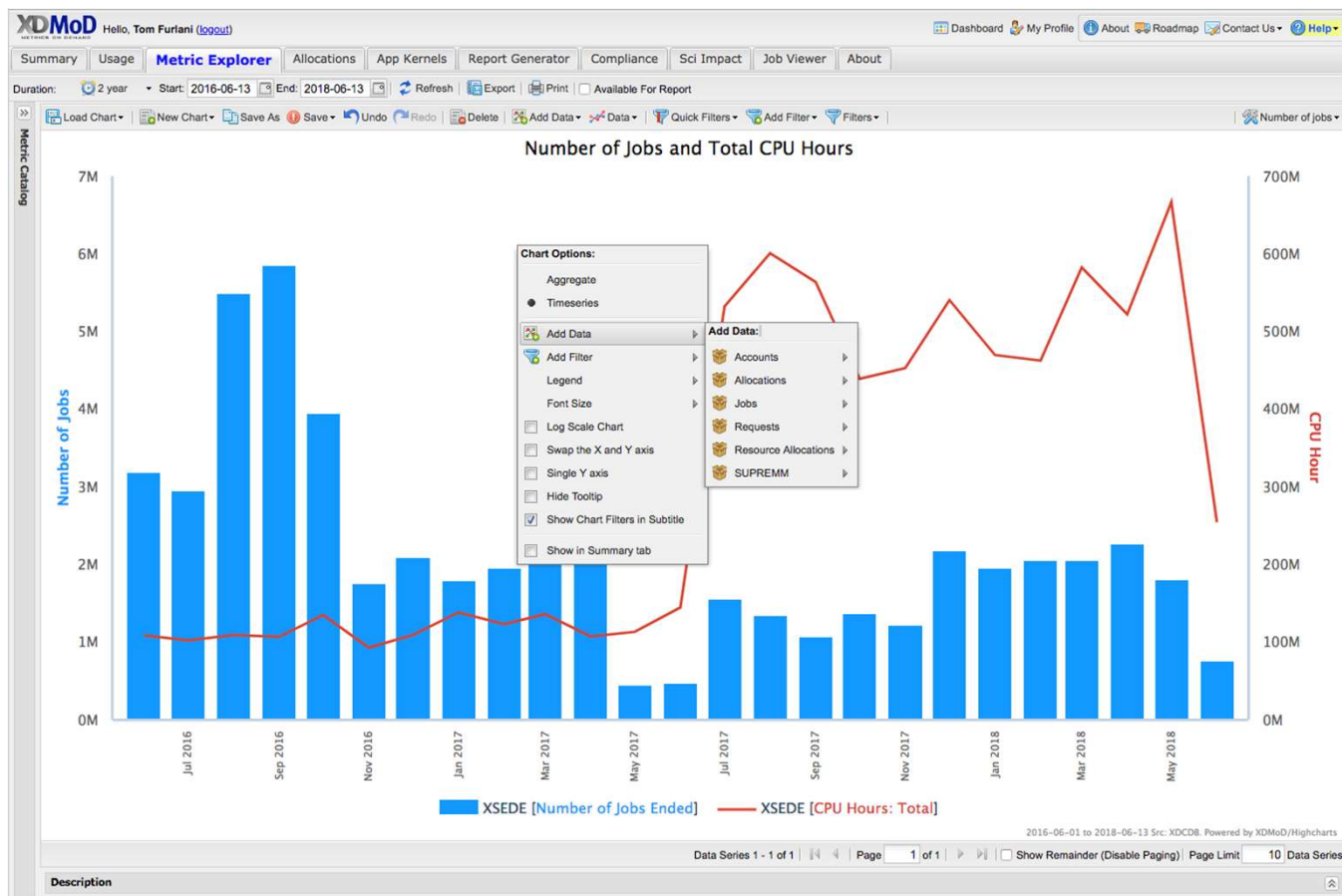






# XDMoD Portal

- **Web-based**
  - Point and click drill down capability
- **Display metrics**
  - Utilization, performance, scientific impact
- **Role based access**
  - User
  - Principal Investigator
  - Center Staff
  - Center Director
- **Custom Report Builder**

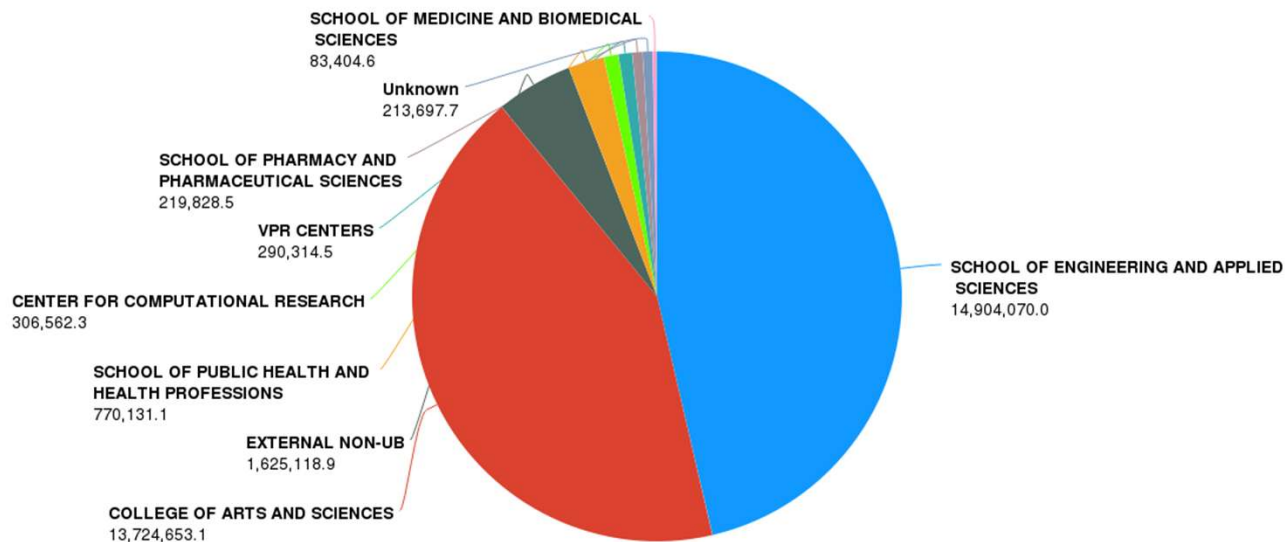




# Easily Obtain Utilization Metrics

CPU hours consumed by campus units

CPU Hours Consumed by Decanal Unit  
Resource = ub-hpc



SCHOOL OF ENGINEERING AND APPLIED SCIENCES COLLEGE OF ARTS AND SCIENCES EXTERNAL NON-UB  
SCHOOL OF PUBLIC HEALTH AND HEALTH PROFESSIONS CENTER FOR COMPUTATIONAL RESEARCH VPR CENTERS  
SCHOOL OF PHARMACY AND PHARMACEUTICAL SCIENCES Unknown SCHOOL OF MEDICINE AND BIOMEDICAL SCIENCES  
ROSWELL PARK INSTITUTE

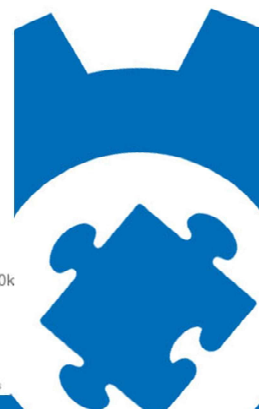
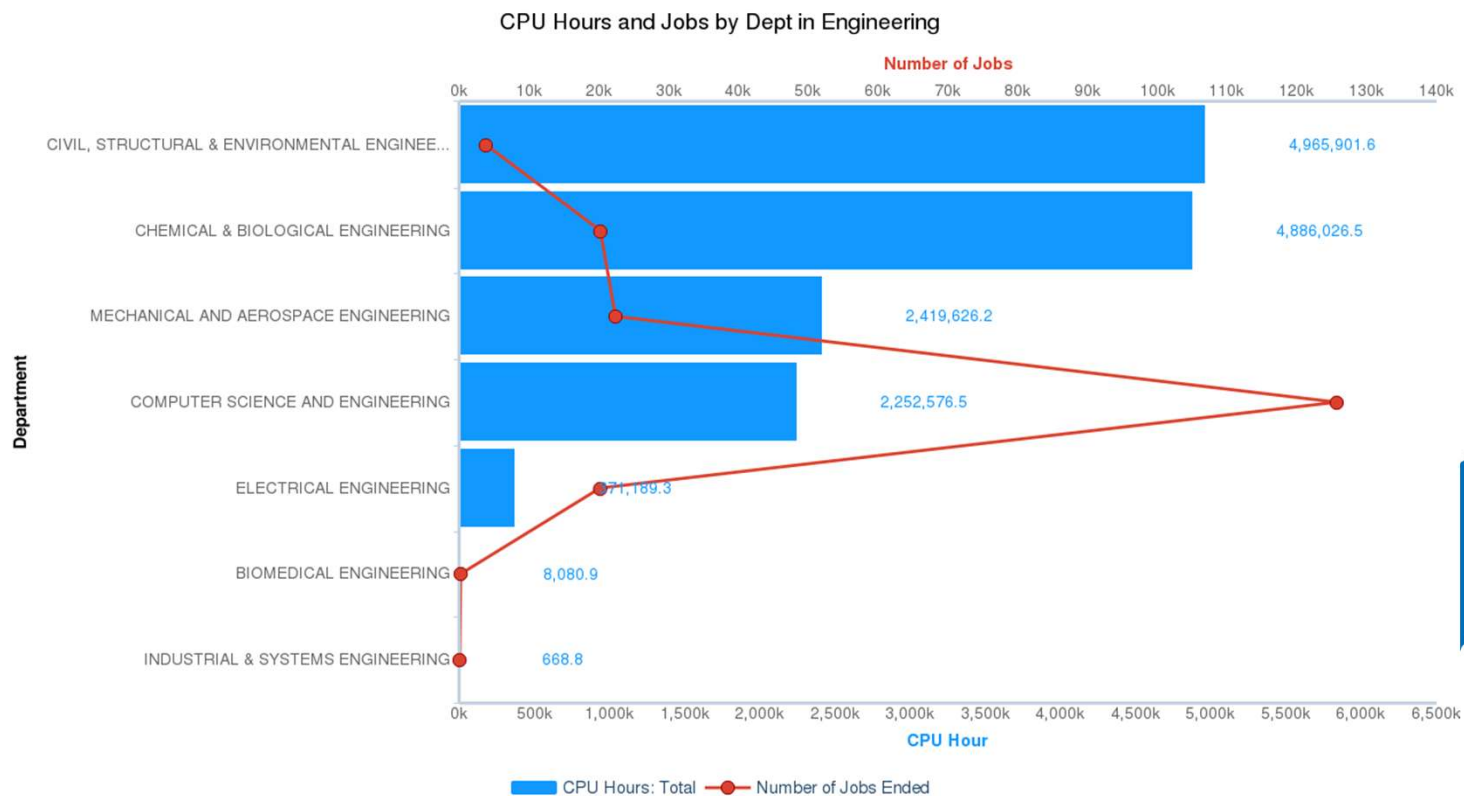






# Drill Down for Greater Detail

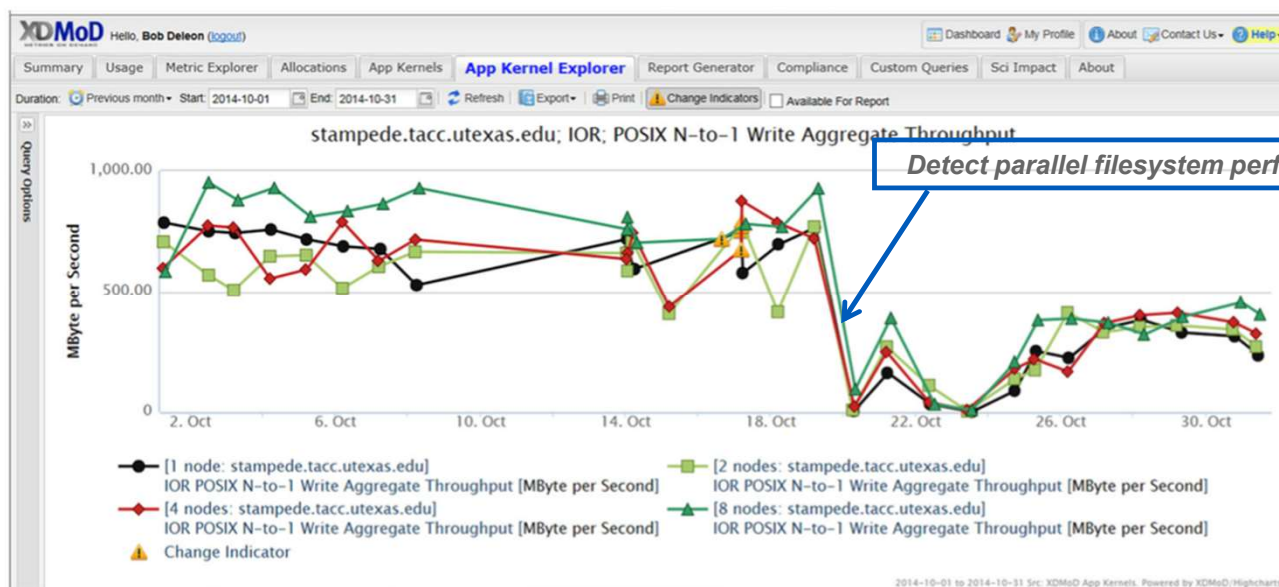
## CPU hours and jobs by Engineering Department





## QoS: Application Kernels

- **Computationally lightweight**
  - Run continuously and on demand to actively measure performance
- **Measure system performance from User's perspective**
  - Local scratch, global filesystem performance, local processor-memory bandwidth, allocatable shared memory, processing speed, network latency and bandwidth
- **Proactively identify underperforming hardware and software**





## Measuring Job Level Performance

- Detailed job-level performance data
- Run on every cluster node and collects hardware counter data
  - CPU usage, Memory usage, I/O usage
- Identify poorly performing jobs (users) and applications
  - Support personnel/Users can use tools to identify/diagnose problems

“Traffic Lights”  
Indicate poor core utilization



*Job employs 32 cores but only 1 core is active at a time with control swapping between the 32 cores. Minor change to the submission script resulted in an improvement of a factor of 30*





# Open XDMoD - OnDemand Integration

Provide XDMoD information directly to OnDemand users

University at Buffalo  
Center for Computational Research

OnDemand provides an integrated, single access point for all of your HPC resources.

Pinned Apps A featured subset of all available apps

Active Jobs  
System Installed App

Home Directory  
System Installed App

Academic Cluster Shell Access  
System Installed App

Desktops

Academic Cluster (JB-HPC) Desktop  
System Installed App

Academic Cluster (JB-HPC) Desktop - Advanced Options  
System Installed App

Faculty Cluster Desktop  
System Installed App

Faculty Cluster Desktop - Advanced Options  
System Installed App

RPC Visualization Desktop  
System Installed App

Via Node - CUDA Desktop  
System Installed App

Via Node - OpenGL Desktop  
System Installed App

Via Node - OpenGL DISABLED Desktop  
System Installed App

GUIs

MATLAB  
System Installed App

Notebooks

Jupyter Notebook - Academic Cluster  
System Installed App

Jupyter Notebook - Faculty Cluster  
System Installed App

Jobs Efficiency Report - 2021-07-24 to 2021-08-23

98.4% efficient 1.6% inefficient  
7 inefficient jobs/451 total jobs

Core Hours Efficiency Report - 2021-07-24 to 2021-08-23

99.4% efficient 0.6% inefficient  
2314.8 inefficient core hours/362312.7 total core hours

Recently Completed Jobs - 2021-07-24 to 2021-08-23

| ID      | Name        | Date | CPU  |
|---------|-------------|------|------|
| 6425757 | SAD         | 8/1  | 97.4 |
| 6420719 | SAD         | 8/7  | 95.8 |
| 6411077 | SAD         | 8/6  | 94.8 |
| 6406852 | FISBATCH    | 8/5  | 93.9 |
| 6403459 | SAD         | 8/5  | 94.7 |
| 7482208 | interactive | 8/4  | 90.0 |
| 6394589 | SAD         | 8/4  | 94.4 |
| 7382205 | hpcg        | 8/3  | 95.9 |
| 6384857 | SAD         | 8/3  | 92.4 |
| 6378820 | SAD         | 8/2  | 91.3 |

Showing first 10 of 451 jobs. See your Open XDMoD dashboard for more information.



Can access job viewer in a single click

→ Overall job efficiency summary

→ Core hours efficiency summary

→ Individual job level efficiency







# Open XDMoD - OnDemand Integration

Provide XDMoD information directly to OnDemand users

Open OnDemand / Job Composer Jobs Templates

## Jobs

Optional configuration for Job Composer adds a direct link to XDMoD job viewer for each job and a customizable message to users about job data availability

[+ New Job](#)

[☆ Create Template](#)

[Edit Files](#) [Job Options](#) [Open Terminal](#) [Submit](#) [Stop](#) [Delete](#)

Show  entries

Search:

| Created               | Name                    | ID                                                                                                                       | Cluster     | Status    |
|-----------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| March 12, 2025 8:38am | Basic Python Serial Job | <a href="#">24 - Open XDMoD</a><br>This job may not appear in Open XDMoD until 24 hours after the completion of the job. | HPC Cluster | Completed |

Showing 1 to 1 of 1 entries

[Previous](#) [1](#) [Next](#)



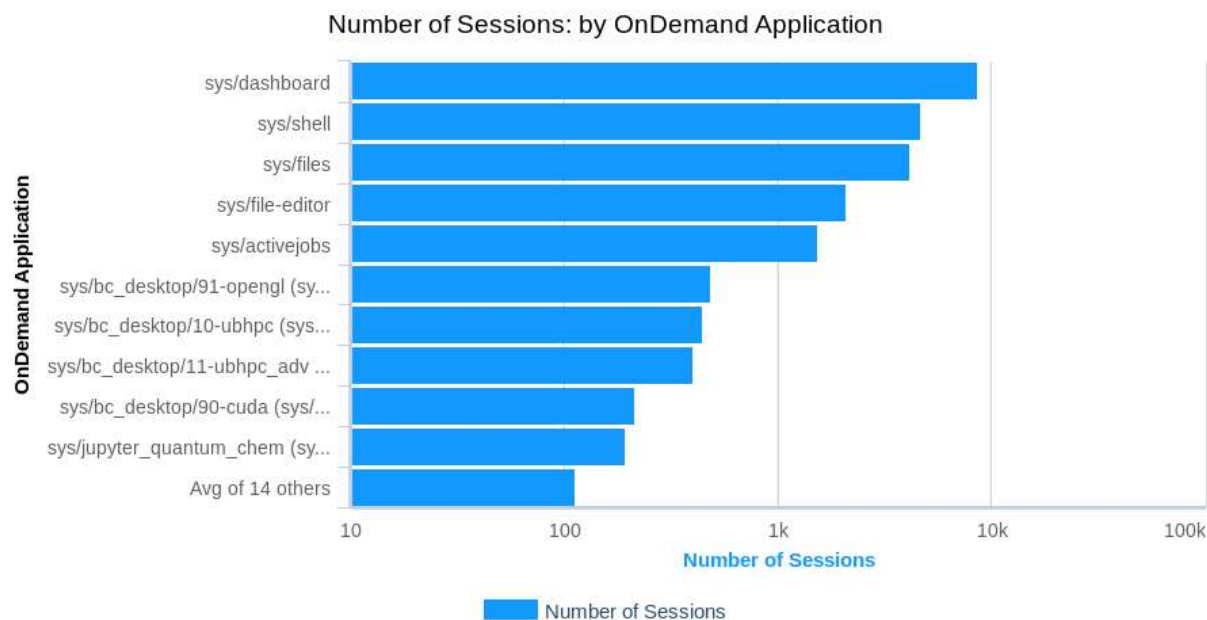




# OnDemand – Open XDMoD Integration

## View OnDemand usage in XDMoD

- Who is using OnDemand?
- What are they running?
- Where are they running from?



2021-01-01 to 2021-12-31 Src: Open OnDemand server logs. Powered by XDMoD/Highcharts





**Ohio Supercomputer Center**  
An **OH·TECH** Consortium Member

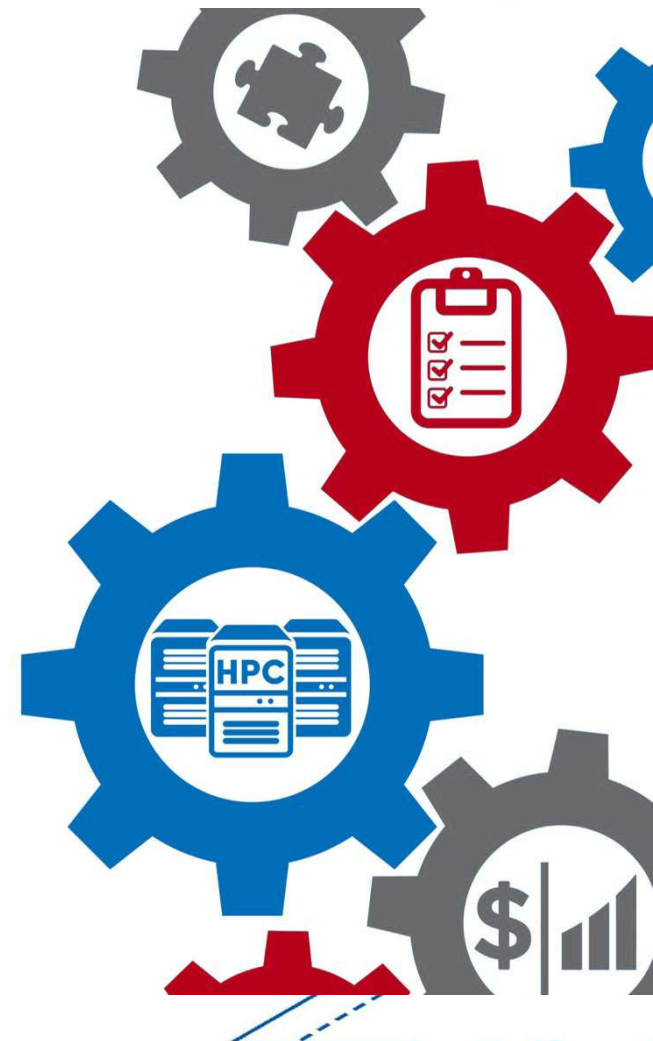


University at Buffalo

Center for Computational Research

## ColdFront – Managing Access

- Used as the source of record in an HPC center to ensure security & continuity of the systems
- Provides center staff ability to manage center resources & who has access to them
- Portal for users to manage their access to center resources & report on their research
- Plug-ins for job scheduler (Slurm), central authentication, job statistics (XDMoD), OnDemand, that enable automation of access to or removal from resources
- Reports for center management to demonstrate the center's impact (publications, grants, research output)





**Ohio Supercomputer Center**  
An **OH·TECH** Consortium Member



University at Buffalo

Center for Computational Research

## Contributing Tutorial & Product Dev Staff:

Andrew Bruno, UB

Alan Chalker, OSC

Andrew Collins, OSC

Robert DeLeon, UB

Trey Dockendorf, OSC

David Hudak, OSC

Matt Jones, UB

Jeff Ohrstrom, OSC

Ryan Rathsam, UB

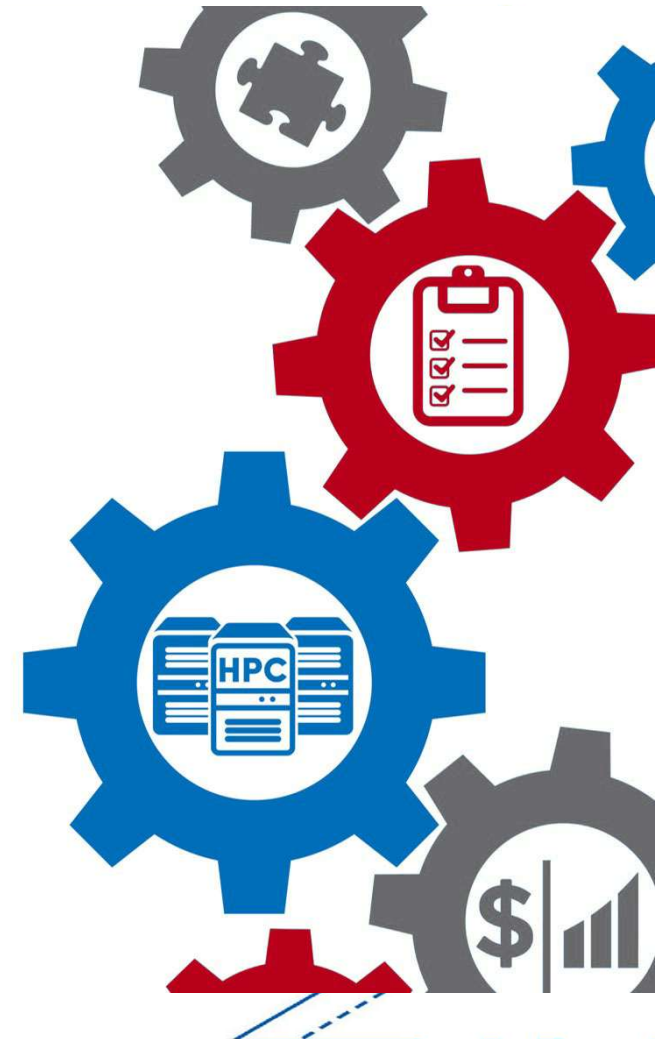
Conner Saeli, UB

Andrew Stoltman, UB

Aaron Weedman, UB

Joseph White, UB

Huston Rogers, Mississippi State





**Ohio Supercomputer Center**  
An OH·TECH Consortium Member



University at Buffalo

Center for Computational Research

## Funding and other acknowledgements:

- OnDemand is/has been supported by the National Science Foundation – award numbers [NSF#2303692](#), [NSF#2411375](#), [NSF#1534949](#) and [NSF#1935725](#)
- Open XDMoD is/has been supported by the National Science Foundation – award numbers – [OAC 2137603](#), [ACI 1025159](#) and [ACI 1445806](#) and [OAC 2137603](#)
- We gratefully acknowledge the partnership with [Virginia Tech](#) on the previous OnDemand2.0 NSF project





**Ohio Supercomputer Center**  
An OH·TECH Consortium Member



University at Buffalo

Center for Computational Research

## How to reach us:

Center for Computational Research – <https://buffalo.edu/ccr>

Open XDMoD - <https://open.xdmod.org/>

ColdFront – <https://coldfront.io>

## Connect with us!

Join the Slack organization for the tutorial

<https://tinyurl.com/hpctoolset>

▼ Channels

# coldfront

# general

# ondemand

🔒 tutorial\_staff

# xdmod